



# SAFETY DATA SHEET

This safety data sheet complies with the requirements of:  
SS586: 2008 (2014)

Revision date 23-Aug-2022

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** ReadyPrep 2-D Rehydration/Sample Buffer 1

### Other means of identification

**Catalogue Number(s)** 1632083, 10009795

**UN proper shipping name** ENVIRONMENTALLY HAZARDOUS SUBSTANCES, SOLID, N.O.S.

**Description** 3077, ENVIRONMENTALLY HAZARDOUS SUBSTANCES, SOLID, N.O.S. (Thiourea), 9, III

**Pure substance/mixture** Mixture

Contains Thiourea

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

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Thailand

For further information, please contact

**Technical Service** +66 2 652 8313  
ctsthailand@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Singapore: 65-31581349

## SECTION 2: Hazards identification

### GHS Classification

|                                 |            |
|---------------------------------|------------|
| <b>Carcinogenicity</b>          | Category 2 |
| <b>Reproductive toxicity</b>    | Category 2 |
| <b>Chronic aquatic toxicity</b> | Category 2 |

### Label elements

**Signal word**

Warning

**Hazard statements**

H351 - Suspected of causing cancer

H411 - Toxic to aquatic life with long lasting effects

**Precautionary Statements - Prevention**

Obtain special instructions before use

Do not handle until all safety precautions have been read and understood

Avoid release to the environment

Wear protective gloves/protective clothing/eye protection/face protection

**Precautionary Statements - Response**

Collect spillage

**Precautionary Statements - Storage**

Store locked up

**Precautionary Statements - Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other hazards which do not result in classification****SECTION 3: Composition/information on ingredients****Substance**

Not applicable

**Mixture**

| Chemical name | EC No     | CAS No  | Weight-% |
|---------------|-----------|---------|----------|
| Urea          | 200-315-5 | 57-13-6 | 50 - 100 |
| Thiourea      | 200-543-5 | 62-56-6 | 20 - 35  |

Non-hazardous  
ingredients

Proprietary

Balance

**SECTION 4: First aid measures****Description of first aid measures****General advice**

IF exposed or concerned: Get medical advice/attention. Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air.

**Eye contact**

Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.

**Skin contact**

Wash skin with soap and water. In the case of skin irritation or allergic reactions see a physician.

Ingestion Rinse mouth thoroughly with water.

**Most important symptoms and effects, both acute and delayed**

Symptoms No information available.

**For emergency responders**

Self-protection of the first aider No information available.

**Indication of any immediate medical attention and special treatment needed**

Note to physicians Treat symptomatically.

## **SECTION 5: Firefighting measures**

**Suitable Extinguishing Media**

Suitable Extinguishing Media Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media No information available.

**Specific hazards arising from the chemical**

Specific hazards arising from the chemical None known.

**Special protective actions for fire-fighters**

Special protective equipment and precautions for fire-fighters Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## **SECTION 6: Accidental release measures**

**Personal precautions, protective equipment and emergency procedures**

Personal precautions Ensure adequate ventilation.

Other information Refer to protective measures listed in Sections 7 and 8.

For emergency responders Use personal protection recommended in Section 8.

**Environmental precautions**

Environmental precautions See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

Methods for containment Prevent further leakage or spillage if safe to do so.

Methods for cleaning up Pick up and transfer to properly labeled containers.

Prevention of secondary hazards Clean contaminated objects and areas thoroughly observing environmental regulations.

Reference to other sections See section 8 for more information. See section 13 for more information.

## **SECTION 7: Handling and storage**

**Precautions for safe handling**

|                                                                            |                                                                                                                                                                                                               |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b>                                             | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Remove contaminated clothing and shoes. |
| <b>General hygiene considerations</b>                                      | Do not eat, drink or smoke when using this product. Wash hands before breaks and immediately after handling the product.                                                                                      |
| <b><u>Conditions for safe storage, including any incompatibilities</u></b> |                                                                                                                                                                                                               |
| <b>Storage Conditions</b>                                                  | Store locked up.                                                                                                                                                                                              |

## SECTION 8: Exposure controls/personal protection

### Control parameters

|                                     |                                                                                                                                                             |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Occupational exposure limits</b> | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

### **Biological occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

|                             |                                                     |
|-----------------------------|-----------------------------------------------------|
| <b>Engineering controls</b> | Showers<br>Eyewash stations<br>Ventilation systems. |
|-----------------------------|-----------------------------------------------------|

### Individual protection measures, such as personal protective equipment

|                                        |                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>             | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| <b>Skin and body protection</b>        | Wear suitable protective clothing.                                                                                                                                       |
| <b>Hand protection</b>                 | Wear suitable gloves.                                                                                                                                                    |
| <b>Respiratory protection</b>          | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                |

## SECTION 9: Physical and chemical properties

### Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Solid                    |
| <b>Appearance</b>      | solid                    |
| <b>Colour</b>          | white                    |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                       | <u>Values</u>     | <u>Remarks • Method</u> |
|---------------------------------------|-------------------|-------------------------|
| <b>pH</b>                             | 10                |                         |
| <b>Melting point / freezing point</b> | No data available | None known              |
| <b>Boiling point / boiling range</b>  | No data available | None known              |
| <b>Flash point</b>                    | No data available | None known              |
| <b>Evaporation rate</b>               | No data available | None known              |
| <b>Flammability (solid, gas)</b>      | No data available | None known              |
| <b>Flammability Limit in Air</b>      |                   | None known              |

|                                        |                   |            |
|----------------------------------------|-------------------|------------|
| Upper flammability or explosive limits | No data available |            |
| Lower flammability or explosive limits | No data available |            |
| Vapour pressure                        | No data available | None known |
| Vapour density                         | No data available | None known |
| Relative density                       | No data available | None known |
| Water solubility                       | Soluble in water  |            |
| Solubility(ies)                        | No data available | None known |
| Partition coefficient                  | No data available | None known |
| Autoignition temperature               | No data available | None known |
| Decomposition temperature              |                   | None known |
| Kinematic viscosity                    | No data available | None known |
| Dynamic viscosity                      | No data available | None known |

Other information No information available

## SECTION 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

Possibility of hazardous reactions None under normal processing.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials None known based on information supplied.

### Hazardous decomposition products

Hazardous decomposition products None known based on information supplied.

## SECTION 11: Toxicological information

### Information on likely routes of exposure

#### Product Information

Inhalation Specific test data for the substance or mixture is not available.

Eye contact Specific test data for the substance or mixture is not available.

Skin contact Specific test data for the substance or mixture is not available.

Ingestion Specific test data for the substance or mixture is not available.

### Symptoms related to the physical, chemical and toxicological characteristics

**Symptoms** No information available.

### Acute toxicity

#### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 4,300.30 mg/kg  
ATEmix (dermal) 2,889.50 mg/kg

#### Component Information

| Chemical name                                  | Oral LD50            | Dermal LD50          | Inhalation LC50        |
|------------------------------------------------|----------------------|----------------------|------------------------|
| Urea                                           | = 8471 mg/kg ( Rat ) |                      |                        |
| Thiourea                                       | = 1750 mg/kg ( Rat ) | > 6810 mg/kg ( Rat ) | > 0.9 mg/L ( Rat ) 4 h |
| 1,3-Propanediol,<br>2-amino-2-(hydroxymethyl)- | = 5900 mg/kg ( Rat ) | > 5000 mg/kg ( Rat ) |                        |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.

**Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

**Respiratory or skin sensitization** Based on available data, the classification criteria are not met.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

**Carcinogenicity** Contains a known or suspected carcinogen. Classification based on data available for ingredients. Suspected of causing cancer.

| Chemical name | European Union |
|---------------|----------------|
| Thiourea      | Carc. 2        |

**Reproductive toxicity** Contains a known or suspected reproductive toxin. Classification based on data available for ingredients. Suspected of damaging fertility or the unborn child.

| Chemical name | European Union |
|---------------|----------------|
| Thiourea      | Repr. 2        |

**STOT - single exposure** Based on available data, the classification criteria are not met.

**STOT - repeated exposure** Based on available data, the classification criteria are not met.

**Aspiration hazard** Classification not possible.

## **SECTION 12: Ecological information**

### Ecotoxicity

**Ecotoxicity** Toxic to aquatic life. Toxic to aquatic life with long lasting effects.

**Unknown aquatic toxicity** Contains 0 % of components with unknown hazards to the aquatic environment

| Chemical name | Algae/aquatic plants | Fish                                                  | Crustacea                               |
|---------------|----------------------|-------------------------------------------------------|-----------------------------------------|
| Urea          | -                    | LC50: 16200 - 18300mg/L (96h,<br>Poecilia reticulata) | EC50: =3910mg/L (48h,<br>Daphnia magna) |

|          |                                                                                                                    |                                                                                                        |                                            |
|----------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Thiourea | EC50: 3.8 - 10mg/L (72h, <i>Desmodesmus subspicatus</i> )<br>EC50: =6.8mg/L (96h, <i>Desmodesmus subspicatus</i> ) | LC50: =10000mg/L (96h, <i>Brachydanio rerio</i> )<br>LC50: >600mg/L (96h, <i>Pimephales promelas</i> ) | EC50: =35mg/L (48h, <i>Daphnia magna</i> ) |
|----------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------|

**Persistence and degradability**

**Persistence and degradability** No information available.

**Bioaccumulative potential**

**Bioaccumulation** There is no data for this product.

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Urea          | <-1.73                |
| Thiourea      | -0.92                 |

**Mobility**

**Mobility in soil** No information available.

**PBT and vPvB assessment**

| Chemical name | PBT and vPvB assessment                                       |
|---------------|---------------------------------------------------------------|
| Urea          | The substance is not PBT / vPvB PBT assessment does not apply |
| Thiourea      | The substance is not PBT / vPvB                               |

**Other adverse effects**

**Other adverse effects** No information available

**SECTION 13: Disposal considerations****Waste treatment methods**

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**SECTION 14: Transport information****ADR**

UN number or ID number 3077  
 UN proper shipping name ENVIRONMENTALLY HAZARDOUS SUBSTANCES, SOLID, N.O.S.  
 Transport hazard class(es) 9  
 Labels 9  
 Packing group III  
 Classification code M7  
 Tunnel restriction code (-)  
 Special Provisions 274, 335, 601, 375  
 Description 3077, ENVIRONMENTALLY HAZARDOUS SUBSTANCES, SOLID, N.O.S. (Thiourea), 9, III

**IMDG**

UN number or ID number UN3077  
 UN proper shipping name ENVIRONMENTALLY HAZARDOUS SUBSTANCES, SOLID, N.O.S.

|                                                                           |                                                                                                  |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Description</b>                                                        | UN3077, ENVIRONMENTALLY HAZARDOUS SUBSTANCES, SOLID, N.O.S. (Thiourea), 9, III, Marine pollutant |
| <b>Transport hazard class(es)</b>                                         | 9                                                                                                |
| <b>Packing group</b>                                                      | III                                                                                              |
| <b>Marine pollutant</b>                                                   | P                                                                                                |
| <b>Special Provisions</b>                                                 | 274, 335, 966, 967, 969                                                                          |
| <b>EmS-No</b>                                                             | F-A, S-F                                                                                         |
| <b>Transport in bulk according to Annex II of MARPOL and the IBC Code</b> | No information available                                                                         |

**IATA**

|                                   |                                                                                |
|-----------------------------------|--------------------------------------------------------------------------------|
| <b>UN number or ID number</b>     | UN3077                                                                         |
| <b>UN proper shipping name</b>    | Environmentally hazardous substances, solid, n.o.s.                            |
| <b>Description</b>                | UN3077, Environmentally hazardous substances, solid, n.o.s. (Thiourea), 9, III |
| <b>Transport hazard class(es)</b> | 9                                                                              |
| <b>Packing group</b>              | III                                                                            |
| <b>Special Provisions</b>         | A97, A158, A179, A197, A215                                                    |
| <b>ERG Code</b>                   | 9L                                                                             |

**SECTION 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****Singapore****Environmental Public Health Act**

Dispose of waste product or used containers according to local regulations.

**Hazardous Waste (Control of Export, Import and Transit) Act**

It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations. Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

**Maritime and Port Authority of Singapore (Dangerous Goods, Petroleum and Explosives) Regulations**

Regulated. See section 14 for more information.

**Poison**

Verify that licence requirements are met Verify that requirements related to using, handling, and storing substances subject to prohibition, authorisation or restriction are met

| Chemical name | Poison | Poison Schedule Number |
|---------------|--------|------------------------|
| Urea          | X      | First schedule         |

**Workplace Safety and Health Act**

Comply with the health and safety at work laws.

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status



**SECTION 16: Other information****Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AELG(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Label elements**

P273 - Avoid release to the environment  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P391 - Collect spillage  
P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Issuing Date** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 23-Aug-2022

**Revision Note** Reformatted and updated existing information.

**This safety data sheet complies with the requirements of: SS586: 2008 (2014)**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**